Total No. of Questions : 6]

SEAT No. :

P19

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APR-17/BE/Insem.-20
B.E. (Mechanical Engineering)
INDUSTRIAL ENGINEERING
(2012 Pattern) (Elective - III(C)) (Semester - II)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates :

1) ***Answer Q1. or Q2, Q3 or Q4, Q5 or Q6.***
2) ***Figures to the right indicate full marks.***
3) ***Assume suitable data if necessary.***

Q1) a) Define span of control and delegation of authority. **[4]**

b) Define term:productivity and factors affecting productivity. **[6]**

OR

Q2) a) Write short note on : Craig Harris Model. **[4]**

b) Explain Line Organization, line and staff organization and Matrix organization. **[6]**

Q3) a) Explain with example, method study symbols for recording facts. **[6]**

b) Write short note on : SIMO chart. **[4]**

OR

Q4) a) Write short note on 'Multiple Activity Chart' **[6]**

b) Differentiate between time study and work sampling. **[4]**

Q5) a) Explain following terms - predetermined motion time standards. **[5]**

b) Following are the element times of a machining operation. The corresponding rating and relaxation allowances are given in table as below. **[5]**

Element	Observed Time (Min)	Performance Rating	Ralaxation Allowance(%)
1	0.15	80	13
2	0.05	85	13
3	0.55	90	10
4	1.00	95	12
5	0.1	90	13

Calculate normal time, standard time for this job assuming contingency allowance of 3% of normal time.

P.T.O.

OR

Q6) a) What is allowance? Explain different types of allowances with example. **[6]**

b) A work-study sample of a manufacturing activity conducted over a 40-hour period shows that a worker with an 85% rating produced 12 parts. The worker's idle time was 10% and the allowance factor was 12% **[4]**

Find the normal and standard time for this activity.

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